EXP:1

1.Use Figma to create an interactive diagram illustrating the components of a basic cellular

system.

Aim: -

To create an interactive diagram illustrating the components of a basic cellular system

Procedure: -

1. open Figma

2. Create a new file

3. Select the Frames

4. Fill in the content that is required for presentation

5. Design Visual Elements

6. Make it Interactive

7. Add Annotations and Explanations

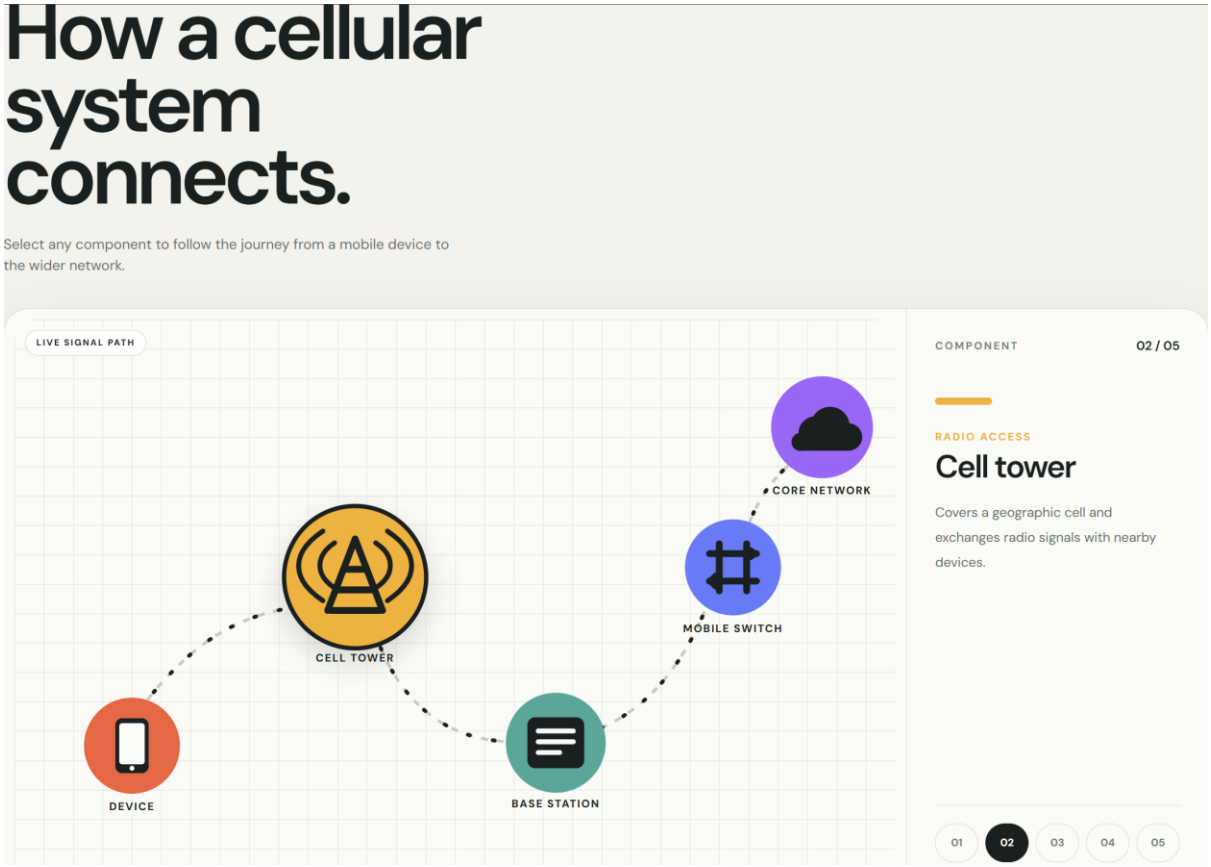
8. Incorporate Multimedia

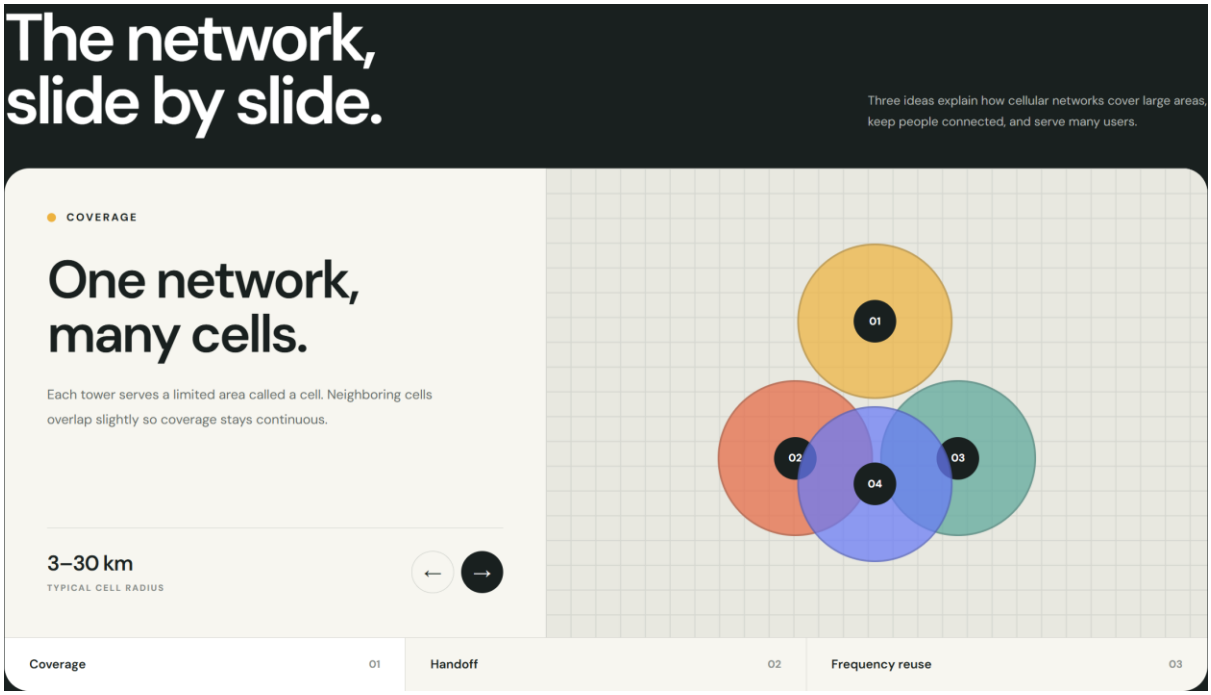
9. Storyboard Animation

11. Review and edit the Prototype

12. Save and Share

Design: -


How a cellular system connects.
Select any component to follow the journey from a mobile device to the wider network.
LIVE SIGNAL PATH
CORE NETWORK
MOBILE SWITCH
CELL TOWER
DEVICE
BASE STATION
COMPONENT
02 / 05
RADIO ACCESS
Cell tower
Covers a geographic cell and exchanges radio signals with nearby devices.
01
02
03
04
05


The network, slide by slide.
Three ideas explain how cellular networks cover large areas, keep people connected, and serve many users.
COVERAGE
One network, many cells.
Each tower serves a limited area called a cell. Neighboring cells overlap slightly so coverage stays continuous.
3–30 km
TYPICAL CELL RADIUS
01
02
03
04
Coverage 01
Handoff 02
Frequency reuse 03

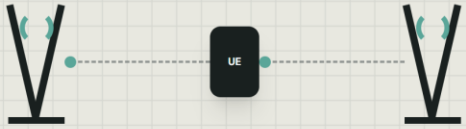
The network, slide by slide.

Three ideas explain how cellular networks cover large areas, keep people connected, and serve many users.

HANDOFF

Connections move with you.

As a device travels, the network measures signal strength and transfers the active connection to a stronger tower.



< 1 sec

TYPICAL HANDOFF TIME

← →

Coverage 01 | Handoff 02 | Frequency reuse 03

The network, slide by slide.

Three ideas explain how cellular networks cover large areas, keep people connected, and serve many users.

FREQUENCY REUSE

Spectrum works harder.

Separated cells can reuse the same radio frequencies, multiplying network capacity without causing interference.



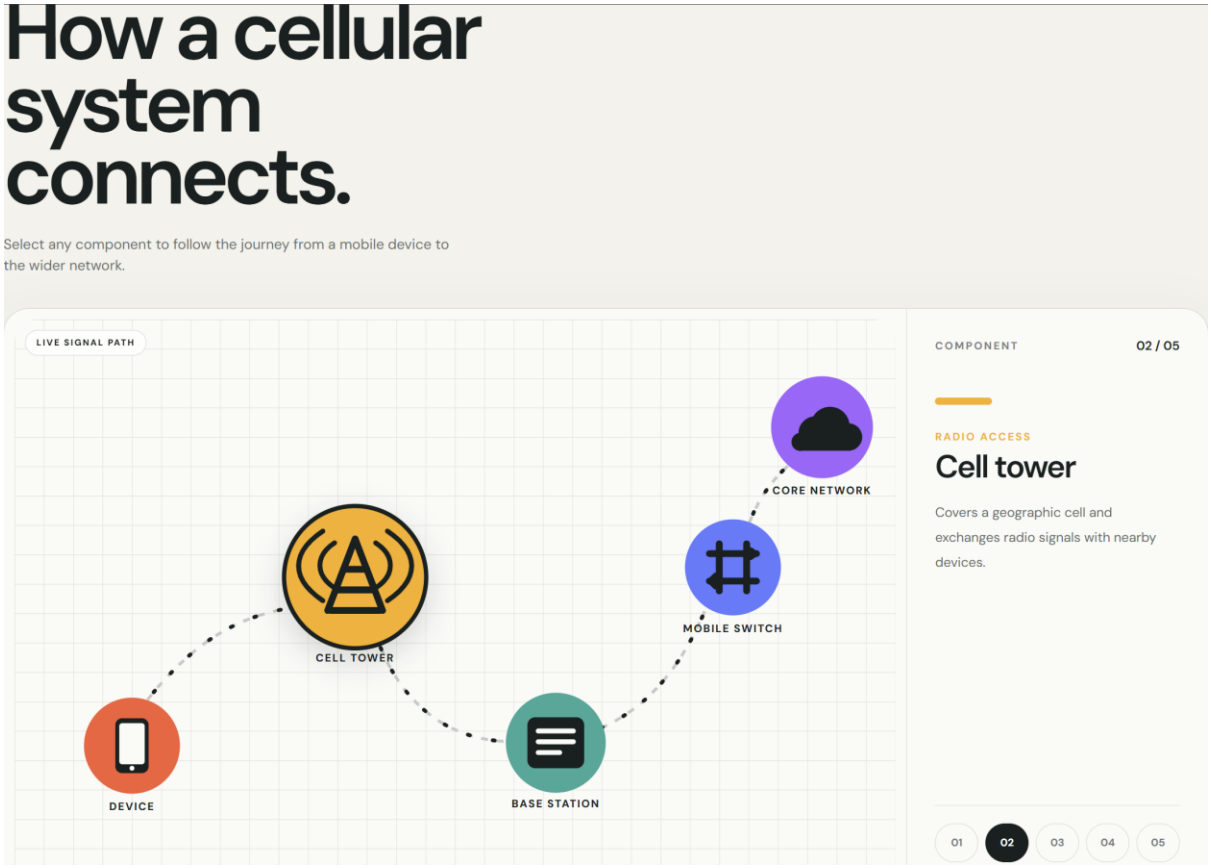
7 cells

CLASSIC REUSE CLUSTER

← →

Coverage 01 | Handoff 02 | Frequency reuse 03

PROTOTYPE:



Result: -

Hence an interactive diagram illustrating the components of a basic cellular system using Figma is created and executed successfully